

Vision-Language Models for Panoramic Dental Radiographs

Evaluation workflow · 4 models × 3 prompting strategies × 3 datasets · 3 research questions

1 · DATA SOURCES

Tufts Dental Database
1,000 panoramic · Universal numbering

DENTEX (QED)
752 panoramic · FDI · 4 disease classes

PR-Reports
121 cases · expert radiologist reports

2 · Ground-truth preparation & verification
32-slot missing-teeth masks (Tufts) · abnormal-tooth + disease labels (DENTEX) · paired expert reports · re-extraction & readability checks

3 · Model zoo × prompting strategies — common inference interface

LLaVA-1.5-7B
general-purpose VLM

LLaVA-Med-v1.5-7B
medical VLM

HuatuoGPT-Vision-7B
medical VLM

DentVLM
dental-specialised VLM

Prompting:

Zero-shot

Few-shot ★

Chain-of-thought

 ★ strongest strategy overall

4 · Inference harness
Slurm jobs on Deakin GPU host · 2× RTX 4000 Ada (20 GB) · fp16 · per-model checkpointing · logged & reproducible → raw generations

5 · Schema-constrained parser
Strict output schema · explicit numbering (Universal ↔ FDI) · empty / off-schema output kept as all-negative → per-tooth predictions

6 · SCORING & METRICS

Patient-level classification
Macro-F1 · accuracy · specificity
balanced acc. · 95% patient bootstrap

Multi-label pathology
Hamming · example / label F1 · Jaccard
Macro-4 F1 vs no-finding baseline

Report quality (RQ3)
BLEU-4 · ROUGE-1/2/L · BERTScore-F1
KL & JS divergence · Likert (planned)

Hallucination & error-type audit
CHR = 1 – precision · hallucinations / image · correct-condition-wrong-tooth vs unsupported localisation · image-level bootstrap (seed 42)

7 · RESEARCH QUESTIONS

RQ1 · Diagnostic performance
how do general / medical / dental VLMs compare?

RQ2 · Prompting strategies
does zero / few / CoT change reliability?

RQ3 · Report quality
how close are reports to expert references?

8 · Outputs & synthesis
Summary tables · per-class & strategy figures · CHR & divergence plots · per-case JSON → research analysis & manuscript
Headline: medical instruction tuning > dental branding · few-shot best · disease-level F1 near floor · CHR ≈ 0.9 on DENTEX

Analysis · Discussion · Publication